

Microeconomics II – Class 17

Consumer's intertemporal choice



Class summary

- * Intertemporal choice and inflation
 - * the budget constraint and inflation
 - * nominal and real interest rates
- * Net present value
 - * growth rates
 - * net present value vs. internal rate of return
 - * net present value and the budget constraint
- * Generalization of the budget constraint
 - * with n periods
 - * with n periods and variable interest rate
- * Bonds
 - * perpetual bonds
- * Readings: Varian (cap. 10)



Intertemporal choice and inflation

- * Goal: to write down the budget constraint in the presence of inflation
- * Price vector: (p_1, p_2)
- * Monetary value of the endowment: $(p_1 m_1, p_2 m_2)$
- * What is the monetary value of the feasible consumption in period 2?

$$p_2 \times c_2 = p_2 \times m_2 + (1+r) \times p_1 \times (m_1 - c_1) \Leftrightarrow$$
$$\Leftrightarrow c_2 = m_2 + \frac{(1+r)}{p_2} \times p_1 \times (m_1 - c_1)$$



Intertemporal choice and inflation

- * Goal: to write down the budget constraint in the presence of inflation
- * Price variation: $p_2 - p_1$
- * Inflation rate (if price increases) / deflation rate (if price decreases): $\frac{p_2 - p_1}{p_1}$

$$\pi = \frac{p_2 - p_1}{p_1} \Leftrightarrow$$

$$\Leftrightarrow \pi \times p_1 = p_2 - p_1 \Leftrightarrow \pi \times p_1 + p_1 = p_2 \Leftrightarrow$$

$$\Leftrightarrow p_2 = (1 + \pi) \times p_1$$



Intertemporal choice and inflation

- * Goal: to write down the budget constraint in the presence of inflation

- * Substituting in the budget constraint:

$$\begin{cases} c_2 = m_2 + \frac{(1+r)}{p_2} \times p_1 \times (m_1 - c_1) \\ p_2 = (1+\pi) \times p_1 \end{cases} \Leftrightarrow$$

$$\Leftrightarrow c_2 = m_2 + \frac{(1+r)}{(1+\pi)} \times (m_1 - c_1)$$

Slope of the budget constraint in the presence of inflation:

$$\frac{\partial c_2}{\partial c_1} = -\frac{(1+r)}{(1+\pi)}$$



Real interest rate

- * $\rho \equiv$ real interest rate (after taking inflation into account)

$$1 + \rho = \frac{(1+r)}{(1+\pi)} \Leftrightarrow \rho = \frac{(1+r)}{(1+\pi)} - 1 \Leftrightarrow \rho = \frac{(1+r) - (1+\pi)}{(1+\pi)} \Leftrightarrow \rho = \frac{(r - \pi)}{(1+\pi)}$$

- * For a low inflation rate, $r - \pi$ is a good approximation of the real interest rate.
- * Slope of the budget constraint in the presence of inflation: $-(1 + \rho)$



Real interest rate

- * Example:

- * Nominal interest rate: $r = 0,1$ (10%)

- * Inflation rate: $\pi = 0,05$ (5%)

- * Goal: to calculate the real interest rate

- * Answer:

$$\rho = \frac{1 + 0,1}{1 + 0,05} - 1 \approx 0,048 \approx 0,05 = r - \pi$$



Real interest rate

$$c_2 = m_2 + \frac{(1+r)}{(1+\pi)} \times (m_1 - c_1) \Leftrightarrow c_2 = m_2 + (1+\rho) \times (m_1 - c_1)$$

* For $\pi = 0$ we have: $c_2 = m_2 + (1+r) \times (m_1 - c_1)$

* Summary:

* Budget constraint without inflation: $c_2 = m_2 + (1+r) \times (m_1 - c_1)$

* Budget constraint with inflation: $c_2 = m_2 + (1+\rho) \times (m_1 - c_1)$



Net Present Value (NPV)

- * Future value in period t : FV or V_t
- * Present value in period 0 : PV or V_0

- * **Compounding**

- * In discrete time: $V_t = V_0 \times (1 + r)^t$
- * In continuous time: $V_t = V_0 \times e^{rt}$

$$V_t = V_0 \times (1 + r)^t \Leftrightarrow V_0 = \frac{V_t}{(1 + r)^t}$$

- * **Discounting:**

- * In discrete time: $V_0 = V_t \times (1 + r)^{-t}$
- * In continuous time: $V_t = V_0 \times e^{-rt}$

$$V_t = V_0 \times e^{rt} \Leftrightarrow V_0 = V_t \times e^{-rt}$$

- * **Discounting:**

- * In discrete time: $(1 + r)^{-t}$
- * In continuous time: e^{-rt}



Growth rates

- * Discrete case

$$V_t = V_0(1+r)^t$$

$$g_V \equiv \frac{\Delta V}{V_t} \equiv \frac{V_t - V_{t-1}}{V_{t-1}} = \frac{V_0(1+r)^t - V_0(1+r)^{t-1}}{V_0(1+r)^{t-1}} = 1 + r - 1 = r$$

- * Continuous case

$$V_t = V_0 e^{rt}$$

$$g_V \equiv \frac{\dot{V}}{V_t} \equiv \frac{\frac{dV_t}{dt}}{V} = \frac{V_0 e^{rt} r}{V_0 e^{rt}} = r$$



Net Present Value (NPV)

- * How do we determine how many years does it take for compounding to increase the value to a given figure?
- * Example: with an interest rate of 5%, how many years does it take for the initial value to double?

$$V_t = 2V_0; r = 0,05$$

$$V_t = 2V_0 \Leftrightarrow V_0(1+r)^t = 2V_0 \Leftrightarrow V_0(1+0,05)^t = 2V_0 \Leftrightarrow (1,05)^t = 2 \Leftrightarrow$$

- * How to isolate the “t”? Hint: use logarithms.

$$\Leftrightarrow \ln(1,05)^t = \ln 2 \Leftrightarrow t \ln 1,05 = \ln 2 \Leftrightarrow t = \frac{\ln 2}{\ln 1,05} \Leftrightarrow t = 14,21 \text{ years}$$



Net Present Value (NPV)

- * Benefit in period t : B_t
- * Cost in period t : C_t
- * Net benefit in period t : $NB_t = B_t - C_t$

- * Net Present Value:
$$NPV = \sum_{t=0}^T \frac{NB_t}{(1+r)^t} = \sum_{t=0}^T \frac{B_t - C_t}{(1+r)^t}$$

- * Decision rule:
 - * Do projects with $NPV \geq 0$

- * Alternative measure: cost-benefit ratio
 - * (do the project if ≥ 1)

$$\frac{\sum_{t=0}^T \frac{B_t}{(1+r)^t}}{\sum_{t=0}^T \frac{C_t}{(1+r)^t}}$$



Internal Rate of Return (IRR)

- * For a predetermined series of cash-flows (or, in this case, net benefits), the result depends on the interest/discount rate considered.
 - * The impact of a change in r on NPV will depend on the sign of each net benefit and the periods in which they occur.
 - * Example: An investment project with a large initial investment (cost) and a series of positive future cash-flows (benefits) will see its NPV decrease if the interest rate rises.
- * In alternative to NPV , we can calculate the Internal Rate of Return (IRR), which is the interest rate that makes $NPV = 0$.

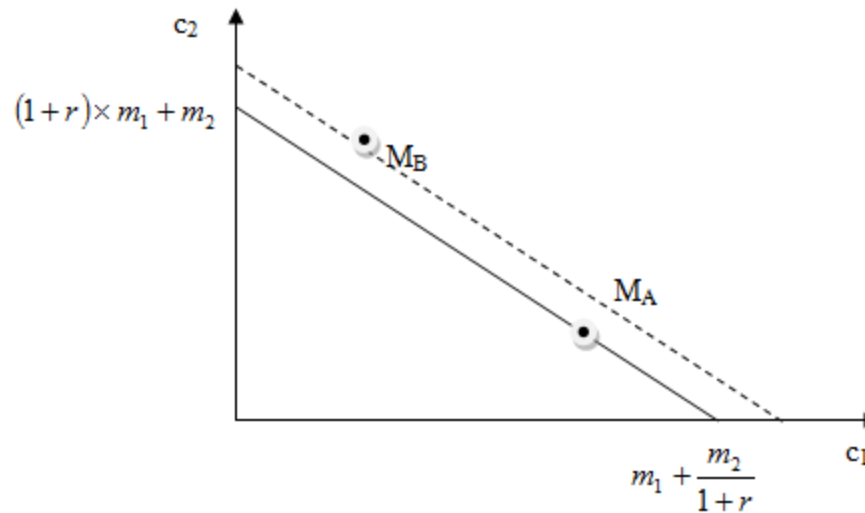
$$NPV = 0 \Leftrightarrow \sum_{t=0}^T \frac{NB_t}{(1 + IRR)^t} = 0 \Leftrightarrow IRR = \dots$$

- * Decision rule: do projects for which $IRR \geq r$



Net present value and the budget constraint

- * Which of the following endowments is preferred, M_A or M_B ?

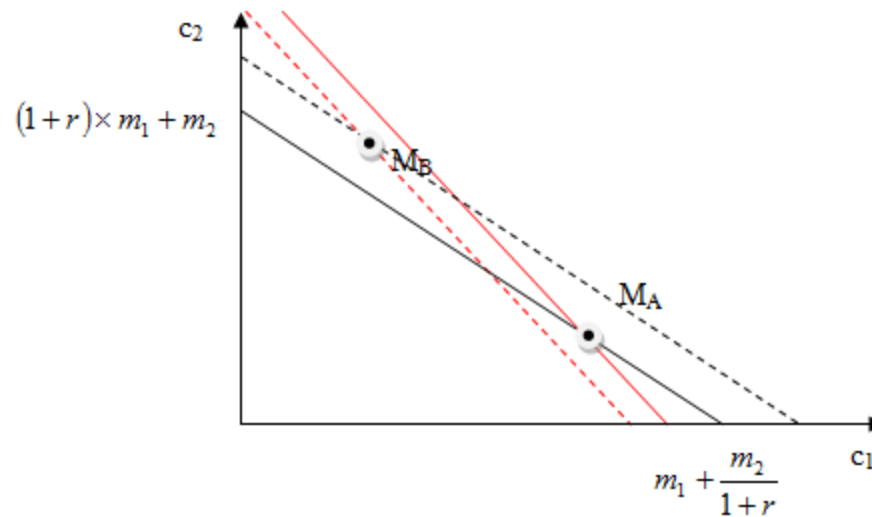


Answer: M_B , at the current interest rate (r , which creates the slope of the budget constraint represented, $-(1+r)$) M_B has a larger net present value, therefore it expands consumption possibilities



Net present value and the budget constraint

- * Which of the following endowments is preferred, M_A or M_B ?



For different interest rates the answer could differ!



Generalizations of the budget constraint

* With n periods

$$c_1 + \frac{c_2}{(1+r)} + \frac{c_3}{(1+r)^2} + \dots + \frac{c_n}{(1+r)^{n-1}} = m_1 + \frac{m_2}{(1+r)} + \frac{m_3}{(1+r)^2} + \dots + \frac{m_n}{(1+r)^{n-1}} \Leftrightarrow$$
$$\Leftrightarrow \sum_{t=1}^n \frac{c_t}{(1+r)^{t-1}} = \sum_{t=1}^n \frac{m_t}{(1+r)^{t-1}}$$

or

$$c_1 - m_1 + \frac{c_2 - m_2}{(1+r)} + \frac{c_3 - m_3}{(1+r)^2} + \dots + \frac{c_n - m_n}{(1+r)^{n-1}} = 0 \Leftrightarrow$$
$$\Leftrightarrow \sum_{t=1}^n \frac{c_t - m_t}{(1+r)^{t-1}} = 0$$

Note: If we had used 0 instead of 1 for the first period, there would be no discrepancy between the number for the year and the power in the denominator.



Bonds

- * Bonds:

- * Bonds are a security (financial instrument with a promised payment schedule) which, in return for an initial investment, pays a regular fixed amount (coupon) and a final fixed amount (face value) at the maturity date.
- * Bonds are a way of borrowing (as opposed to equity/stock which make their buyer an owner of the issuing firm).

- * Calculating the present value of a bond:

$$PV = \frac{x}{(1+r)} + \frac{x}{(1+r)^2} + \dots + \frac{x+F}{(1+r)^T} = \sum_{t=1}^T \frac{x}{(1+r)^t} + \frac{F}{(1+r)^T}$$

- * Notation:

x: Coupon paid regularly
F: Face value
T: Maturity



Bonds

- * An increase in the interest rate decreases the bond's present value.
 - * if a bond pays a fixed interest of 2% regularly and the market interest rate increases from 1% to 3%, what happens to the investor's willingness to buy the bond?
- * For each value that the market interest rate may assume, there will be a bond price (V) that makes the net present value of investing in the bond null

$$NPV = 0 \Leftrightarrow \sum_{t=1}^T \frac{x}{(1+r)^t} + \frac{F}{(1+r)^T} - V = 0 \Leftrightarrow V = \sum_{t=1}^T \frac{x}{(1+r)^t} + \frac{F}{(1+r)^T} \Leftrightarrow V = PV$$



Consols or perpetuities

- * A perpetuity is a bond that pays coupons forever.
- * Present value of a perpetuity (alternative way to calculate the PV)

$$PV = \sum_{t=1}^{\infty} \frac{x}{(1+r)^t} = \frac{x}{(1+r)} \left[\sum_{t=1}^{\infty} \frac{1}{(1+r)^{t-1}} \right] = \frac{x}{(1+r)} \left[\frac{1}{1 - \left(\frac{1}{1+r} \right)} \right] = \frac{x}{(1+r)} \left(\frac{1+r}{1+r-1} \right) = \frac{x}{r}$$

Note: Sum of a converging geometric sequence: $1 + \delta + \delta^2 + \dots = \sum_{t=1}^{\infty} \delta^{t-1} = \frac{1}{1-\delta} \quad \forall 0 \leq \delta < 1$



Consols or perpetuities

- * Alternative reasoning: How much would we have to invest today (V) at the interest rate r to have a fixed periodic remuneration of x ?

$$Vr = x \Leftrightarrow V = \frac{x}{r}$$

